# Computing Distance and Midpoint from Coordinates

High School Geometry

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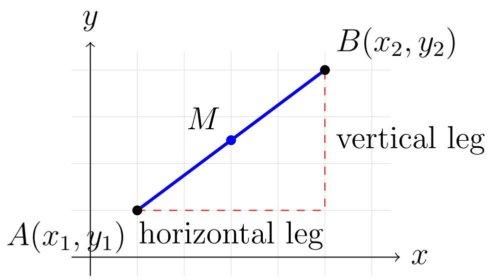
**Where the two formulas come from.** Drop a horizontal leg and a vertical leg from the segment's endpoints. The segment is the hypotenuse of a right triangle whose legs are $|x_2 - x_1|$ and $|y_2 - y_1|$, so the Pythagorean theorem gives the length. The midpoint is not a length at all — it is the *average* of the two endpoints, taken one coordinate at a time.




*How every segment on this sheet is labelled: $A(x_1, y_1)$ and $B(x_2, y_2)$ are the endpoints and $M$ is the midpoint. No values are shown here — use the coordinates each problem gives you.*

- Distance: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Midpoint: $M = \left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2}\right)$
- Leave a distance in exact radical form unless the problem says otherwise; show the substitution step before you simplify.

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**1.** Find the distance between $A(1, 2)$ and $B(4, 6)$. Show the substitution into the distance formula, and name the two legs of the right triangle you are using.

Answer: ________________

**2.** Find the midpoint of the segment joining $A(2,-3)$ and $B(8,7)$. Write it as an ordered pair.

Answer: ________________

**3.** Find the exact distance between $A(-3,1)$ and $B(2,13)$. Watch the signs when you subtract the negative $x$-coordinate.

Answer: ________________

**4.** Find the midpoint of $A(-5,4)$ and $B(3,-2)$. Then state which quadrant the midpoint lies in.

Answer: ________________

**5.** Find the distance between the origin $A(0, 0)$ and $B(6, 6)$. Give the answer in exact radical form, fully simplified — not a rounded decimal.

Answer: ________________

**6.** $M(3, 5)$ is the midpoint of $\overline{AB}$, and one endpoint is $A(-1, 2)$. Find the coordinates of $B$.
(Think of the midpoint formula as two averaging equations, one for $x$ and one for $y$, and solve each for the missing coordinate. Then check your $B$ by recomputing the midpoint.)

Answer: ________________

**7.** Triangle $PQR$ has vertices $P(1, 1)$, $Q(7, 1)$, and $R(4, 6)$.

(a) Find the midpoint $M$ of side $\overline{PQ}$.

Answer: ________________

(b) Find the length of the median $\overline{RM}$ — the segment from $R$ to that midpoint.

Answer: ________________

**8. Coordinate proof.** Quadrilateral $ABCD$ has vertices $A(0,0)$, $B(6,2)$, $C(8,7)$, and $D(2,5)$. Its diagonals are $\overline{AC}$ and $\overline{BD}$.
Compute the midpoint of each diagonal. Then write a short proof: state what you found and explain why sharing a midpoint proves that the diagonals **bisect each other**. Two or three complete sentences.